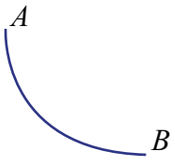


The shortest distance path between *A* and *B*

Fig. 11.1 (a)



The shortest time route between *A* and *B*

Fig. 11.1 (b)

Certainly this route is not the shortest time route joining the points A and B, because the velocity of the motion in the straight line (Fig. 11.1(a)) will be comparatively slow; whereas it take a curve that is steeper near A (Fig. 11.1(b)), even though the path becomes longer, a considerable portion of the distance will be covered at a greater speed. The solution to this problem is solved by **Integral calculus.** This is called Brachistochrone problem which initiates the study of calculus of variation using integral tool.

Situation 2

In elementary geometry we have learnt to evaluate the measure of the following regular shape of the figures given below by using known formulae.

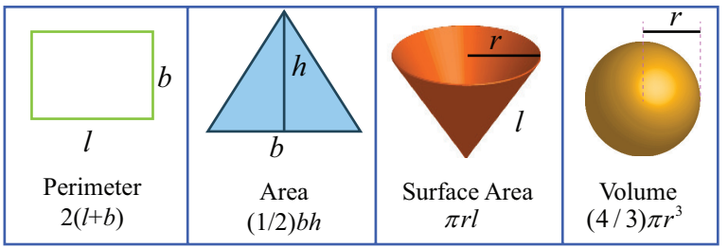


Fig. 11.2 (a)

How can the measure of the following figures given by functions be calculated?

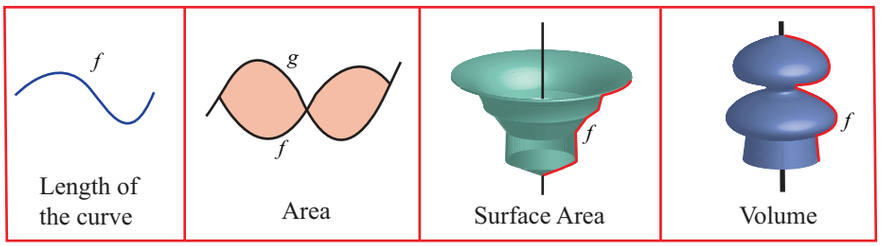


Fig. 11.2 (b)

Though the problems look so difficult, integral calculus solves it without any difficulties.

Situation 3

At a particular moment, a student needs to stop his speedy bike to avoid a collision with the barrier ahead at a distance 40 metres away from him. Immediately he slows (acceleration) down the bike applying brake at a rate of 8 meter/second^2. If the bike is moving at a speed of 24m/s, when the brakes are applied, will it stop before collision?

Also look at the following problems that occur naturally in our life.



Fig. 11.3

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